

Quantum Portfolio CoPilot

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Project

Classical vs Quantum Portfolio Optimization

Built using:

- Python
- Qiskit
- Streamlit
- Plotly

Project Statistics

Total Assets

30

Quantum Assets

15

Selected Assets

4

Quantum Algorithm

QAOA

Quantum Portfolio CoPilot

Classical Portfolio Optimization vs Quantum Portfolio Optimization using QAOA

This dashboard compares the performance of a **Classical Portfolio Optimizer** and a **Quantum Portfolio Optimizer (QAOA)**.

It provides portfolio metrics, comparison charts, and selected asset allocations to demonstrate the application of quantum computing in financial optimization.

> About this Project

Portfolio Metrics

Expected Return

0.1286

↑ +0.0414 vs Classical

Portfolio Risk

0.2105

↑ +0.0471 vs Classical

Sharpe Ratio

0.6111

↑ +0.0773

Classical Assets

8

Quantum Assets

4

Performance Summary

Quantum achieved a higher expected return.

Quantum has a better Sharpe Ratio.

Classical portfolio has lower risk.

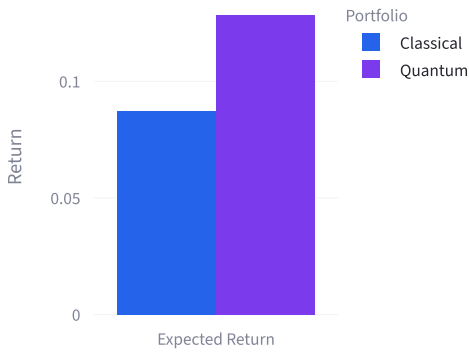
Classical portfolio is more diversified.

Comparison Results

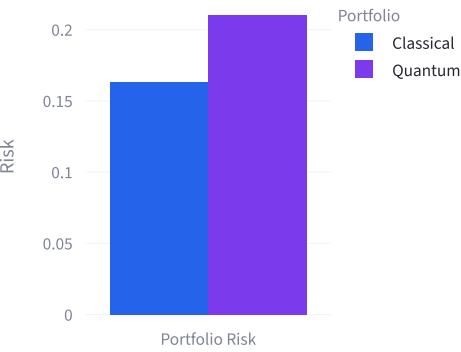
	Metric	Classical	Quantum
0	Expected Return	0.0873	0.1287
1	Portfolio Risk	0.1635	0.2105
2	Sharpe Ratio	0.5338	0.6111
3	Selected Assets	8	4

Portfolio Comparison Charts

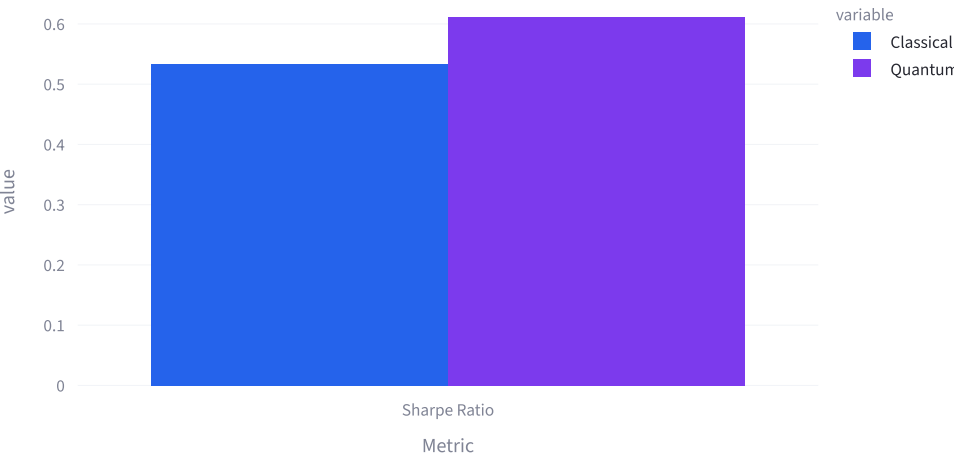
Expected Return Comparison



Portfolio Risk Comparison



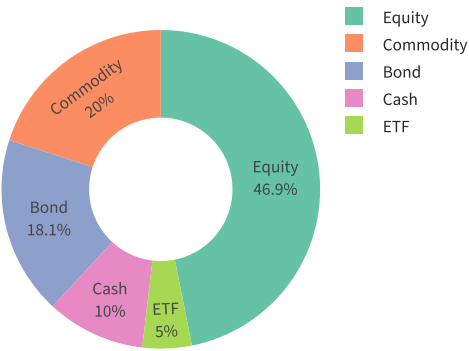
Sharpe Ratio Comparison



Portfolio Allocation

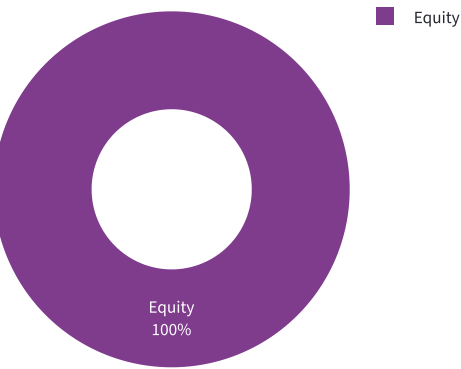
Classical Portfolio

Classical Asset Allocation



Quantum Portfolio

Quantum Asset Allocation



 Classical Portfolio

	Asset_ID	Asset_Name	Asset_Class	Ex
0	A12	Equity_12	Equity	
1	A05	Equity_5	Equity	
2	A24	Commodity_24	Commodity	
3	A13	Bond_13	Bond	
4	A29	Cash_29	Cash	
5	A23	ETF_23	ETF	
6	A03	Equity_3	Equity	
7	A09	Equity_9	Equity	

 Quantum Portfolio

	Asset_ID	Asset_Name	Asset_Class	Expe
0	A05	Equity_5	Equity	
1	A06	Equity_6	Equity	
2	A08	Equity_8	Equity	
3	A12	Equity_12	Equity	

 Download Project Results

- ↓ Classical Portfolio
- ↓ Quantum Portfolio
- ↓ Comparison Results

 Quantum Portfolio CoPilot

Version 2.0

Developed by Navadhikannan

Built with Python • Qiskit • Streamlit • Plotly